Speaker Notes — Guyton & Hall, Chapter 59

The Limbic System and the Hypothalamus — Behavioral and Motivational Mechanisms of the Brain
Course: English for Medicine II
Presenters: Hassan Ghaffari (7:40), Bardia Asadi (7:15), Arash Behrad (7:35), MohammadReza Movahed (7:35) — total 30:05, timed at about 110 words per minute plus a short pause at each click

How to use these notes

- `click` means: press the right arrow or space bar. Each click shows the next bullet or the next part of a diagram, together with any red or green outline on the figure. Say the sentences that follow the `click` while the new part appears.
- The same text, with the same `click` cues, is in each slide's speaker notes, so it shows in Presenter View.
- Key terms show how to say the hard words, in brackets. To read them: read each letter the simple way: **a** as in car, **e** as in bed, **i** as in sit, **o** as in go, **ee** = long i (see), **oo** = long u (food), **ai** as in my, **ay** as in day; the **bold** part is said a little stronger. Example: hypothalamus (hai-po-**tha**-la-mus).
- **Quick Check** slides: the answer is **not** on the slide, and there is nothing to click. Read the question and the three options, wait about 4 seconds (count silently), then say the answer yourself. You may point at someone and ask "What do you think?" The pause is already in the slide time.
- **Prediction** slides (slides 25 and 34): ask the question, wait about 4 seconds, then click to show the result.
- **Questions at the end:** the last slide is "Thank You! Questions?". Stay on it while you answer questions.
- Use this file to rehearse. During the talk, use the short Cue Notes (same slides, same clicks). Rehearse until the cues alone are enough, so you can look at the audience instead of reading.
- If you speak slower (about 95 words per minute), each part takes about one minute longer. That is still inside the 5–10 minute limit.

Hassan Ghaffari — Section 1: Activating–Driving Systems of the Brain

Slides 1–11 · total time 460 seconds (7:40)

Slide 1 — The Limbic System and the Hypothalamus (title slide)

What to say:

Hello, everyone. My name is Hassan Ghaffari, and my teammates are Bardia Asadi, Arash Behrad and MohammadReza Movahed. Today we will present Chapter 59 of Guyton and Hall: The Limbic System and the Hypothalamus, Behavioral and Motivational Mechanisms of the Brain. In other words: what keeps the brain awake, and what drives our behavior.

Key terms to stress: limbic system (**lim**-bik); hypothalamus (hai-po-**tha**-la-mus); motivational (mo-ti-**vay**-sho-nal)

Time: 30 seconds

Slide 2 — What Does Your Brain Keep? (opening question)

What to say:

Let's start with a question. Think about your way to class this morning. How much of it do you remember? Probably very little. In fact, Guyton and Hall say the brain usually throws away more than 99 percent of the information it could learn. `click` So what decides the 1 percent we keep? Please keep this question in mind. We will answer it later in our talk.

Key terms to stress: information (in-for-**may**-shen)

Time: 35 seconds · 1 click

Slide 3 — Overview

What to say:

Our talk has four parts. First, I will explain how the brain stem keeps the brain active. Then Bardia will introduce the limbic system and the hypothalamus. Arash will talk about hormones, the body's clock, and behavior. Finally, MohammadReza will explain the hippocampus, the amygdala and the limbic cortex. Each of us will also ask you a question, so please stay awake!

Key terms to stress: brain stem; hippocampus (hi-po-**kam**-pus); amygdala (a-**mig**-da-la)

Time: 35 seconds

Slide 4 — Section 1: Activating–Driving Systems of the Brain (section divider)

What to say:

Let's begin with section one: the activating and driving systems of the brain.

Key terms to stress: activating–driving systems

Time: 5 seconds

Slide 5 — The Brain Stem Drives the Cerebrum (diagram)

What to say:

The cerebrum needs a continuous flow of nerve signals from the lower brain. Without these signals, it becomes useless. click For example, severe compression of the brain stem, sometimes from a pineal tumor, can cause a coma that lasts for the rest of the person's life. The brain stem activates the cerebrum in two ways. click First, it directly stimulates background activity in wide areas of the brain. click Second, it activates neurohormonal systems, which release excitatory or inhibitory transmitters into selected areas.

Key terms to stress: cerebrum (se-**ree**-brum); pineal (**pi**-ni-al); coma (**ko**-ma); neurohormonal (noo-ro-hor-**mo**-nal)

Time: 45 seconds · 3 clicks

Slide 6 — The Reticular Excitatory Area

What to say:

Figure 59-1 shows the main driver of brain activity. click It is an excitatory area in the reticular substance of the pons and mesencephalon, the pink area outlined in red. click It is also called the bulboreticular facilitory area; here, facilitory means excitatory. click Downward, it sends signals to the spinal cord to keep tone in the antigravity muscles. click Upward, most signals go first to the thalamus, which then excites the whole cerebral cortex. Some upward signals are fast: they use acetylcholine and last milliseconds. Others are slow: they build up over seconds and set the brain's background excitability.

Key terms to stress: reticular (re-**ti**-kyu-ler); pons (ponz); mesencephalon (me-zen-**se**-fa-lon); bulboreticular facilitory area (bal-bo-re-**ti**-kyu-ler fe-**si**-li-to-ree); thalamus (**tha**-la-mus); acetylcholine (a-see-til-**ko**-leen)

Time: 55 seconds · 4 clicks

Slide 7 — What Keeps the Excitatory Area Active (diagram)

What to say:

What keeps this excitatory area active? click Mainly, sensory signals that enter the brain from the periphery. Pain signals are especially strong; they excite the brain to attention. click We can see this when the brain stem is cut above the fifth cranial nerves. Almost all sensory input is lost, and the brain falls close to a permanent coma. click The cortex also sends feedback signals back to this area, which then excites the cortex even more. This positive feedback leads to an awake mind.

Key terms to stress: periphery (pe-**ri**-fe-ree); fifth cranial nerves (**kray**-ni-al); positive feedback

Time: 50 seconds · 3 clicks

Slide 8 — Thalamus and Reticular Inhibitory Area

What to say:

Let's look at Figure 59-1 again. click The thalamus, the black circle at the top, is a distribution center. Almost every area of the cortex connects with its own specific area in the thalamus. click Signals also reverberate, which means they travel back and forth, between the thalamus and the cortex. This may help establish long-term memories. click At the bottom is the reticular inhibitory area, in the medulla. click It can inhibit the excitatory area. One way is by exciting neurons that secrete serotonin, an inhibitory neurohormone.

Key terms to stress: thalamus (**tha**-la-mus); reverberate (ri-**ver**-be-rayt); medulla (me-**da**-la); serotonin (se-ro-**to**-nin)

Time: 50 seconds · 4 clicks

Slide 9 — Neurohormonal Control of Brain Activity

What to say:

click The second way is neurohormonal control. Neurohormones are excitatory or inhibitory transmitter agents, and their effects often last for minutes or hours. click Figure 59-2 shows three such systems, mapped in the rat brain. click Norepinephrine is usually excitatory, and it spreads to virtually every area of the brain. click Dopamine goes mainly to the basal ganglia. There, it probably acts as an inhibitory transmitter, but in some other areas it may be excitatory. click Serotonin is usually inhibitory, and it goes mostly to the midline structures.

Key terms to stress: neurohormones (**noo**-ro-**hor**-monz); norepinephrine (nor-e-pi-**ne**-frin); dopamine (**do**-pa-meen); serotonin (se-ro-**to**-nin); basal ganglia (**bay**-sal **gan**-gli-ya)

Time: 50 seconds · 5 clicks

Slide 10 — Four Neurohormonal Systems in Humans

What to say:

Figure 59-3 shows the centers of four such systems in the human brain stem. click The locus ceruleus secretes norepinephrine. It generally excites the brain and probably plays a role in dreaming and REM, or rapid eye movement, sleep. click The substantia nigra sends dopamine to the caudate nucleus and putamen. click The raphe nuclei secrete serotonin, which suppresses pain in the spinal cord and helps cause normal sleep. click Gigantocellular neurons, or giant cells, secrete acetylcholine and make the nervous system awake and excited. click And here is a link to the clinic: destruction of the dopamine neurons in the substantia nigra is the basic cause of Parkinson's disease.

Key terms to stress: locus ceruleus (**lo**-kus se-**roo**-li-yus); substantia nigra (sab-**stan**-shi-ya **nai**-gra); caudate (**ko**-dayt); putamen (pyoo-**tay**-men); raphe nuclei (**ray**-fee **noo**-kli-ai); gigantocellular (jai-gan-to-**se**-lyu-ler); Parkinson's disease (**par**-kin-sonz)

Time: 60 seconds · 5 clicks

Slide 11 — Quick Check: A Case (question for the audience, answer not on the slide)

What to say:

Here is a case for you. A pineal tumor compresses the brain stem where it joins the cerebrum, and the patient falls into a coma that does not end. Why? Is it A, B or C? Think for a moment. (Wait.) The answer is B. Without continuous signals from the brain stem, the cerebrum becomes useless. Now I will hand over to Bardia, who will introduce the limbic system and the hypothalamus.

Key terms to stress: pineal (**pi**-ni-al); compresses (kom-**pre**-siz); coma (**ko**-ma)

Time: 45 seconds

Hassan Ghaffari total: 30 + 35 + 35 + 5 + 45 + 55 + 50 + 50 + 50 + 60 + 45 = 460 seconds (7:40)

Possible audience questions — Hassan Ghaffari

1. **Q:** Why does cutting the brain stem above the fifth cranial nerves cause near coma, but cutting below them does not?
   A: The fifth cranial nerves are the highest nerves that bring many somatosensory signals into the brain. Cutting below them leaves sensory input from the face and mouth, and this input keeps the excitatory area active.
2. **Q:** Are norepinephrine, dopamine, serotonin and acetylcholine the only neurohormones in the brain?
   A: No. The chapter lists many others, such as enkephalins, gamma-aminobutyric acid, glutamate, histamine and endorphins. Each system plays its own role in controlling a different quality of brain function.

Bardia Asadi — Section 2: The Limbic System and the Hypothalamus

Slides 12–20 · total time 435 seconds (7:15)

Slide 12 — Section 2: The Limbic System and the Hypothalamus (section divider)

What to say:

Thank you, Hassan. In section two, I will introduce the limbic system and its control center, the hypothalamus.

Key terms to stress: limbic system (**lim**-bik); hypothalamus (hai-po-**tha**-la-mus)

Time: 10 seconds

Slide 13 — What Is the Limbic System?

What to say:

click The limbic system is the entire neuronal circuitry that controls emotional behavior and motivational drives. click The word limbic means border. At first, it described the border structures around the basal regions of the cerebrum. click A major part of the limbic system is the hypothalamus. click Besides behavior, these areas control internal conditions of the body, such as body temperature and the drives to eat and drink. These are called vegetative functions.

Key terms to stress: limbic (**lim**-bik); neuronal circuitry (**noo**-ro-nal **ser**-ki-tree); motivational drives; vegetative functions (**ve**-je-tay-tiv)

Time: 40 seconds · 4 clicks

Slide 14 — Functional Anatomy of the Limbic System

What to say:

Figure 59-4 shows the anatomy of the limbic system: an interconnected complex of basal brain structures. click In the middle lies the extremely small hypothalamus. click Around it are other deep structures, such as the septum, the anterior nucleus of the thalamus, the hippocampus and the amygdala. click Around these deep parts is the limbic cortex, a ring of cortex on each side of the brain. It starts in the orbitofrontal area and the subcallosal gyrus, and ends in the parahippocampal gyrus and uncus. click On top, the ring passes over the corpus callosum as the cingulate gyrus.

Key terms to stress: septum (**sep**-tum); hippocampus (hi-po-**kam**-pus); amygdala (a-**mig**-da-la); orbitofrontal (or-bi-to-**fran**-tal); parahippocampal gyrus (pa-ra-hi-po-**kam**-pal **jai**-rus); uncus (**an**-kus); corpus callosum (**kor**-pus ka-**lo**-sum); cingulate gyrus (**sin**-gyu-let)

Time: 55 seconds · 4 clicks

Slide 15 — Key Position of the Hypothalamus

What to say:

Figure 59-5 shows the key position of the hypothalamus, right in the middle of the limbic system. click It weighs only about 4 grams, less than 1 percent of the brain mass. click Still, it has two-way pathways with all levels of the limbic system. An important route is the medial forebrain bundle, a two-way trunk line to the brain stem. click It sends signals in three directions. Down to the brain stem and autonomic nerves. Up to the anterior thalamus and limbic cortex. And into the infundibulum, to the pituitary gland. click So the hypothalamus controls most vegetative and endocrine functions of the body.

Key terms to stress: autonomic (o-to-**no**-mik); infundibulum (in-fan-**di**-byu-lum); pituitary (pi-**too**-i-te-ree); medial forebrain bundle (**mee**-di-al); endocrine (**en**-do-krin)

Time: 60 seconds · 4 clicks

Slide 16 — Blood Pressure and Body Temperature

What to say:

Now, what does the hypothalamus control? Figure 59-6 shows its control centers. click Stimulating the posterior and lateral hypothalamus raises the arterial pressure and the heart rate. click Stimulating the preoptic area often does the opposite: it lowers both. These effects are sent mainly through cardiovascular control centers in the pons and medulla. click The anterior hypothalamus, especially the preoptic area, regulates body temperature. click It has temperature-sensitive neurons. When warmer blood flows through this area, they become more active. When the blood is cooler, they become less active. These neurons then control mechanisms that raise or lower body temperature.

Key terms to stress: arterial (ar-**tee**-ri-yal); posterior (pos-**tee**-ri-yer); preoptic (pree-**op**-tik); cardiovascular (kar-di-yo-**vas**-kyu-ler); temperature-sensitive neurons

Time: 55 seconds · 4 clicks

Slide 17 — Body Water: Thirst and Antidiuretic Hormone (diagram)

What to say:

The hypothalamus controls body water in two ways. Both start when the body fluids become too concentrated. click The first way is thirst. A thirst center lies in the lateral hypothalamus. When the fluid there becomes too concentrated, we feel an intense desire to drink, and we drink until it is normal again. click The second way is control of water loss in the urine, mainly by the supraoptic nuclei. Their nerve fibers run down to the posterior pituitary, where the endings secrete antidiuretic hormone, also called vasopressin. click This hormone travels in the blood to the kidneys, which reabsorb more water. So the body fluids move back toward normal.

Key terms to stress: thirst center; supraoptic nuclei (soo-pra-**op**-tik **noo**-kli-ai); posterior pituitary (pi-**too**-i-te-ree); antidiuretic hormone (an-tee-dai-yu-**re**-tik); vasopressin (vay-zo-**pre**-sin); reabsorb (ree-ab-**zorb**)

Time: 60 seconds · 3 clicks

Slide 18 — Oxytocin: Labor and Milk Ejection (diagram)

What to say:

When the paraventricular nuclei are stimulated, their neurons secrete the hormone oxytocin. click Oxytocin increases the contractility of the uterus, which means the uterus contracts more strongly. click At the end of pregnancy, especially large amounts are secreted, and this helps the labor contractions that expel the baby. click Later, whenever the baby suckles, a reflex signal travels from the nipple to the posterior hypothalamus, and oxytocin is released again. click Now it contracts the ductules of the breast, so milk is expelled through the nipples and the baby can nourish itself.

Key terms to stress: oxytocin (ok-si-**to**-sin); paraventricular (pa-ra-ven-**tri**-kyu-ler); uterine (**yoo**-te-rin); contractility (kon-trak-**ti**-li-tee); suckle (**sa**-kel); ductules (**dak**-tyools)

Time: 50 seconds · 4 clicks

Slide 19 — Hunger and Satiety

What to say:

Figure 59-6 also shows the feeding centers. click Stimulating several areas causes extreme hunger and a voracious appetite, which means a very strong desire to eat. One of them is the lateral hypothalamic area. click Against it works the satiety center, in the ventromedial nuclei. Satiety means the feeling of being full. When this center is stimulated, an animal that is eating suddenly stops and shows no interest in food. click The arcuate nucleus has at least two types of neurons: some increase appetite, and others decrease it. click And the mammillary bodies help control feeding reflexes, such as licking the lips and swallowing.

Key terms to stress: satiety (sa-**tai**-e-tee); voracious (vo-**ray**-shes); ventromedial (ven-tro-**mee**-di-al); arcuate (**ar**-kyu-it); mammillary bodies (**ma**-mi-le-ree)

Time: 60 seconds · 4 clicks

Slide 20 — Quick Check: A Case (question for the audience, answer not on the slide)

What to say:

Here is my question for you. After an injury to the hypothalamus, a patient's body fluids become too concentrated, but his kidneys keep losing water in the urine. Which nuclei are most likely damaged? A: the arcuate nucleus. B: the ventromedial nuclei. C: the supraoptic nuclei. (Wait.) The answer is C, the supraoptic nuclei. Through antidiuretic hormone, they make the kidneys reabsorb water. Now I will hand over to Arash, who will talk about hormones, clocks and behavior.

Key terms to stress: supraoptic (soo-pra-**op**-tik); arcuate (**ar**-kyu-it); antidiuretic (an-tee-dai-yu-**re**-tik)

Time: 45 seconds

Bardia Asadi total: 10 + 40 + 55 + 60 + 55 + 60 + 50 + 60 + 45 = 435 seconds (7:15)

Possible audience questions — Bardia Asadi

1. **Q:** Why is the hypothalamus so important if it is so small?
   A: It has two-way pathways with all levels of the limbic system. It sends signals down to the brain stem, up to the thalamus and limbic cortex, and into the infundibulum to the pituitary gland.
2. **Q:** Are the hypothalamic centers really as exact as the figure shows?
   A: No. Guyton warns that these areas are not nearly as accurately localized as the figures suggest. The effects may also come from fiber tracts that pass through an area, not from the nucleus itself.

Arash Behrad — Section 3: Hormones, Clocks and Behavior

Slides 21–28 · total time 455 seconds (7:35)

Slide 21 — Section 3: Hormones, Clocks and Behavior (section divider)

What to say:

Thank you, Bardia. In section three, I will talk about hormones, the body's master clock, and behavior.

Key terms to stress: hormones (**hor**-monz); behavior (bi-**hay**-vyer)

Time: 10 seconds

Slide 22 — Hypothalamic Control of the Pituitary (diagram)

What to say:

Bardia showed how nerve fibers reach the posterior pituitary. Now, the anterior pituitary. click It is controlled through the blood. Most of its blood first flows through the lower part of the hypothalamus, where hypothalamic nuclei secrete releasing and inhibitory hormones into it. click These hormones travel to the anterior pituitary and control the release of its own hormones. click The posterior pituitary is different: it is controlled by nerve fibers that run down through the infundibulum. So, one gland, two kinds of control.

Key terms to stress: anterior pituitary (an-**tee**-ri-yer pi-**too**-i-te-ree); releasing hormones; inhibitory hormones (in-**hi**-bi-to-ree); infundibulum (in-fan-**di**-byu-lum)

Time: 45 seconds · 3 clicks

Slide 23 — The Suprachiasmatic Nucleus: Master Clock

What to say:

Figure 59-8 shows the suprachiasmatic nucleus, or SCN. click It contains about 20,000 neurons and lies above the optic chiasm, where the optic nerves cross. It is a master clock: its firing follows a circadian, or 24-hour, rhythm, and it organizes our cycle of sleeping and waking. click Retinal cells with the pigment melanopsin send it signals through the retinohypothalamic tract. So light can reset the clock. click This is why jet lag does not last. After flying across time zones, the clock is eventually synchronized with local day-night time. click And when the SCN is damaged, the circadian sleep-wake rhythm is lost.

Key terms to stress: suprachiasmatic nucleus (soo-pra-kai-yaz-**ma**-tik); optic chiasm (**kai**-a-zem); circadian (ser-**kay**-di-yan); melanopsin (me-la-**nop**-sin); retinohypothalamic tract (re-ti-no-hai-po-tha-**la**-mik); synchronized (**sin**-kro-naizd)

Time: 60 seconds · 4 clicks

Slide 24 — Inside the Clock: Clock Genes (diagram)

What to say:

How does a clock work inside a cell? Biological clocks are found in nearly every tissue and organ, but without signals from the SCN they usually keep their rhythm for only a few days. click Inside, two proteins, CLOCK and BMAL1, bind together and switch on the clock genes PER and the cryptochrome genes CRY. click PER and CRY proteins build up, and then they inhibit CLOCK and BMAL1. So their own production stops. click This on-off cycle normally takes about 24 hours. In 2017, the Nobel Prize went to Hall, Rosbash and Young for discovering these mechanisms in fruit flies. And no, this Jeffrey Hall is not John Hall, the author of our textbook!

Key terms to stress: BMAL1 (**bee**-mal wan); PER (per); cryptochrome (**krip**-to-krom); Nobel Prize (no-**bel**); Rosbash (**roz**-bash)

Time: 65 seconds · 3 clicks

Slide 25 — Stimulation vs. Lesions: Can You Predict? (audience prediction)

What to say:

Now let's compare two areas. click Stimulating the lateral hypothalamus causes thirst and eating, and it raises general activity, sometimes up to rage. Stimulating the ventromedial nuclei does the opposite: satiety, less eating, and tranquility, which means calm. Now predict: what happens if each area is destroyed on both sides? click Bilateral lesions of the lateral hypothalamus decrease drinking and eating almost to zero. This often leads to lethal starvation, and the animal becomes extremely passive. click Bilateral lesions of the ventromedial areas cause excessive drinking and eating. The animal cannot be satiated, so it eventually develops tremendous obesity. It also becomes hyperactive, with frequent bouts of extreme rage. So, lesions usually cause the opposite of stimulation.

Key terms to stress: lesions (**lee**-zhenz); bilateral (bai-**la**-te-ral); tranquility (tran-**kwi**-li-tee); lethal starvation (**lee**-thal); passivity (pa-**si**-vi-tee); obesity (o-**bee**-si-tee)

Time: 70 seconds · 3 clicks

Slide 26 — Reward and Punishment Centers

What to say:

click Some limbic structures decide whether a sensation is pleasant or unpleasant. We call this reward or punishment. click The major reward centers lie along the medial forebrain bundle. The strongest are in the lateral and ventromedial nuclei of the hypothalamus, shown here in green. click The most potent punishment areas are the central gray area around the aqueduct of Sylvius, in the mesencephalon, and the periventricular zones of the hypothalamus. Only the periventricular zone is on this figure, in red. click But wait: on the last slide, the lateral hypothalamus caused rage. How can it be a reward center? In many areas, weak stimulation feels like reward, and strong stimulation feels like punishment. click Stimulating punishment centers can completely inhibit the reward centers. So punishment and fear can take precedence over pleasure and reward.

Key terms to stress: medial forebrain bundle (**mee**-di-al); aqueduct of Sylvius (**a**-kwe-dakt ov **sil**-vi-yus); periventricular (pe-ri-ven-**tri**-kyu-ler); precedence (**pre**-si-dens)

Time: 75 seconds · 5 clicks

Slide 27 — Rage and Placidity

What to say:

click Strong stimulation of the punishment centers causes a pattern called rage. click The animal takes a defense posture, like the cat in this photo. It extends its claws, hisses, spits and growls. Its hair stands up, which is called piloerection, and its pupils dilate. The slightest provocation causes a savage attack. click In a normal animal, rage is held in check mainly by the ventromedial nuclei, and also by the hippocampi and the anterior limbic cortex. click Stimulating the reward centers does the opposite. It causes placidity and tameness, which means calm and gentle behavior. click This matters in medicine. Tranquilizers, such as chlorpromazine, usually inhibit both centers and reduce how strongly an animal reacts. This is probably how they work in psychotic states.

Key terms to stress: piloerection (pai-lo-i-**rek**-shen); provocation (pro-vo-**kay**-shen); hippocampi (hi-po-**kam**-pai); placidity (pla-**si**-di-tee); tranquilizers (**tran**-kwi-lai-zerz); chlorpromazine (klor-**pro**-ma-zeen); psychotic (sai-**ko**-tik)

Time: 70 seconds · 5 clicks

Slide 28 — Reward and Punishment Guide Learning (diagram)

What to say:

Almost everything we do is related to reward and punishment. They also decide what we learn. A new stimulus excites many areas of the cortex. **click** If it brings neither reward nor punishment, the cortical response fades with repetition. This is habituation: the animal learns to ignore it. **click** If it brings reward or punishment, the response grows stronger. This is reinforcement, and it builds strong memory traces. **click** Now, remember Hassan's first question: what decides the 1 percent we keep? The answer is reward and punishment. These centers select what we learn. Now I will hand over to MohammadReza, who will explain the hippocampus, the amygdala and the limbic cortex.

Key terms to stress: habituation (ha-bi-choo-**ay**-shen); reinforcement (ree-in-**fors**-ment); repetition (re-pe-**ti**-shen)

Time: 60 seconds · 3 clicks

Arash Behrad total: 10 + 45 + 60 + 65 + 70 + 75 + 70 + 60 = 455 seconds (7:35)

Possible audience questions — Arash Behrad

1. **Q:** Are there other reward centers besides the hypothalamus?
   A: Yes, less potent ones: in the septum, the amygdala, certain areas of the thalamus and basal ganglia, and down into the basal tegmentum of the mesencephalon.
2. **Q:** Where do the signals from the SCN go?
   A: Its axons go to other regions of the hypothalamus that control daily changes in body temperature, the sleep-wake cycle and hormones. Nervous and hormonal signals then carry the rhythm to other organs and tissues.

MohammadReza Movahed — Section 4: Hippocampus, Amygdala and Limbic Cortex

Slides 29–38 · total time 455 seconds (7:35)

Slide 29 — Section 4: Hippocampus, Amygdala and Limbic Cortex (section divider)

What to say:

Thank you, Arash. In the last section, I will explain three more parts of the limbic system: the hippocampus, the amygdala and the limbic cortex.

Key terms to stress: hippocampus (hi-po-**kam**-pus); amygdala (a-**mig**-da-la); limbic cortex

Time: 15 seconds

Slide 30 — Functions of the Hippocampus

What to say:

click The hippocampus, labeled here at the bottom left, is an elongated part of the cerebral cortex that folds inward into the lateral ventricle. click Almost any sensory experience activates at least part of it, and its stimulation can cause almost any behavior pattern, such as pleasure, rage or passivity. click It sends signals to the anterior thalamus, the hypothalamus and other limbic areas, especially through the fornix, a major pathway. click It can also become hyperexcitable: weak stimuli can cause focal seizures, with hallucinations that the person knows are not real.

Key terms to stress: lateral ventricle (**ven**-tri-kel); fornix (**for**-niks); hyperexcitable (hai-per-ik-**sai**-ta-bel); seizures (**see**-zherz); hallucinations (ha-loo-si-**nay**-shenz)

Time: 50 seconds · 4 clicks

Slide 31 — The Hippocampus and Memory (diagram)

What to say:

In a few patients, parts of both hippocampi were removed to treat epilepsy. This caused anterograde amnesia: new memories could not be formed. click They still recalled old memories, and their short-term memory lasted seconds up to a minute or two. But they learned almost no new verbal information, not even the names of people they met every day. click Why? The hippocampus began as part of the olfactory, or smell, cortex, where it decided if a smell meant food or danger. So it probably became the brain's judge of importance. click If it marks something as important, it seems to make the mind rehearse it over and over, until it becomes long-term memory.

Key terms to stress: anterograde amnesia (**an**-te-ro-grayd am-**nee**-zha); epilepsy (**e**-pi-lep-see); verbal (**ver**-bal); olfactory (ol-**fak**-to-ree); rehearse (ri-**hers**)

Time: 65 seconds · 3 clicks

Slide 32 — The Amygdala: A “Window” on the World (diagram)

What to say:

The amygdala is a group of small nuclei just under the cortex at the front of each temporal lobe. click One part, the corticomedial nuclei, receives olfactory, or smell, signals. In humans, another part, the basolateral nuclei, is much more developed and works in many other behaviors. click The amygdala receives signals from all of the limbic cortex and from the temporal, parietal and occipital lobes, especially the visual and auditory association areas. So it is called the window through which the limbic system sees the person's place in the world. click It sends signals back to these areas, to the hippocampus, septum and thalamus, and especially to the hypothalamus.

Key terms to stress: corticomedial (kor-ti-ko-**mee**-di-al); basolateral (bay-so-**la**-te-ral); parietal (pa-**rai**-e-tal); occipital (ok-**si**-pi-tal); auditory (**o**-di-to-ree)

Time: 60 seconds · 3 clicks

Slide 33 — Stimulating the Amygdala

What to say:

click Stimulating the amygdala can cause almost all the effects of stimulating the hypothalamus, plus some others. click Through the hypothalamus, it can change arterial pressure, heart rate and gut activity, dilate the pupils, and release anterior pituitary hormones, especially gonadotropins and adrenocorticotropic hormone. click It can cause involuntary movements, such as circling, or licking, chewing and swallowing. click Some amygdala nuclei cause rage and fear; others cause reward and pleasure.

Key terms to stress: gonadotropins (go-na-do-**tro**-pinz); adrenocorticotropic (a-dree-no-kor-ti-ko-**tro**-pik); involuntary (in-**vo**-len-te-ree)

Time: 40 seconds · 4 clicks

Slide 34 — What If Both Amygdalas Are Lost?

What to say:

Let's try one more prediction. In a monkey, the front parts of both temporal lobes are destroyed, together with both amygdalas. How will the monkey behave? Take a moment to guess. click The result is called the Klüver-Bucy syndrome. First, the monkey is not afraid of anything. click It has extreme curiosity about everything. click It forgets rapidly. click It tends to put everything in its mouth, and sometimes even tries to eat solid objects. click And it often has a very strong sex drive, even toward the wrong animals. Such lesions are rare in humans, but affected people respond in a similar way. Guyton says the amygdala helps make our behavior appropriate for each occasion. In this monkey, behavior no longer fits.

Key terms to stress: Klüver-Bucy syndrome (**kloo**-ver **byoo**-see); curiosity (kyoo-ri-**yo**-si-tee); rapidly (**ra**-pid-lee); appropriate (a-**pro**-pri-yet)

Time: 75 seconds · 5 clicks

Slide 35 — Functions of the Limbic Cortex (figure and table)

What to say:

The limbic cortex, the ring outlined in red, is the least understood part of the limbic system. It is a transitional zone that passes signals between the rest of the cortex and the limbic system. click So it works as an association area for the control of behavior. click Ablation, or removal, of parts of it on both sides changes behavior. Removing the anterior temporal cortex also damages the amygdalas, so we see the Klüver-Bucy syndrome again. Removing the posterior orbital frontal cortex causes insomnia and restlessness. And removing the anterior cingulate and subcallosal gyri releases the rage centers, so the animal becomes vicious.

Key terms to stress: transitional (tran-**zi**-sho-nal); ablation (ab-**lay**-shen); insomnia (in-**som**-ni-ya); cingulate (**sin**-gyu-let); subcallosal (sab-ka-**lo**-sal); vicious (**vi**-shes)

Time: 60 seconds · 2 clicks

Slide 36 — Quick Check: A Case (question for the audience, answer not on the slide)

What to say:

Our last question for you. After surgery on both hippocampi for epilepsy, a patient meets a new nurse every day. What happens? A: he cannot learn her name. B: he forgets his childhood. C: he cannot keep her words in mind for even a minute. (Wait.) The answer is A. His old memories and short-term memory stay, but new long-term verbal memories do not form.

Key terms to stress: hippocampi (hi-po-**kam**-pai); epilepsy (**e**-pi-lep-see)

Time: 40 seconds

Slide 37 — Key Takeaways (key takeaways)

What to say:

To finish, here are our key takeaways. click First, the brain stem keeps the cerebrum awake and active. click Second, the hypothalamus, only 4 grams, controls most vegetative and endocrine functions. click Third, reward and punishment decide what we do and what we learn. That is why we keep less than 1 percent of information. click Fourth, the hippocampus turns new information into long-term memory. click And finally, the amygdala helps make our behavior appropriate for each occasion.

Key terms to stress: vegetative (**ve**-je-tay-tiv); endocrine (**en**-do-krin); appropriate (a-**pro**-pri-yet)

Time: 45 seconds · 5 clicks

Slide 38 — Thank You! (closing slide, stay here during questions)

What to say:

Thank you for listening. We are happy to answer your questions.

Key terms to stress:

Time: 5 seconds

MohammadReza Movahed total: 15 + 50 + 65 + 60 + 40 + 75 + 60 + 40 + 45 + 5 = 455 seconds (7:35)

Possible audience questions — MohammadReza Movahed

1. **Q:** Why is the hippocampus so easily excited?
   A: It has a different type of cortex from the rest of the cerebrum. Some of its areas have only three nerve cell layers instead of the six layers found elsewhere, and this is probably one reason.
2. **Q:** What does the rest of the limbic cortex do?
   A: It works as an association area for behavior. For example, the anterior temporal cortex has gustatory and olfactory associations, and the parahippocampal gyri have complex auditory and thought associations.

Reference: Hall, J. E., & Hall, M. E. (2021). *Guyton and Hall textbook of medical physiology* (14th ed.). Elsevier.

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